Two-Tier Geodesic Spectrum Structure in Dehn Fillings of $m003$ and $m006$: Lucas Universality and Slope Encoding

Marvin L. Gentry

Independent Researcher, Seattle, WA

drmlgentry@protonmail.com ORCID: 0009-0006-4550-2663

April 2026

Abstract

We study the geodesic length spectra of Dehn fillings of the two smallest-volume orientable cusped hyperbolic 3-manifolds, $m003$ and $m006$. A scan of 65 fillings reveals a sharp two-tier structure. For $n \geq 21$ ($m003$) and $n \geq 23$ ($m006$), $\log n$ appears as a geodesic length in more than 80% of fillings tested (universal geodesics); $n = 18 = L_6$ lies in the transition zone at 79.4%, just below the threshold. For n below these thresholds, geodesic occurrence is filling-dependent and sparse. The universal tier is anchored by Lucas numbers $L_k = \varphi^k + \varphi^{-k}$: the values $n \in \{18, 23, 29, 37, 47\}$ achieve $\geq 81\%$ occurrence. The sparse tier encodes the Dehn surgery coefficient p: fillings with $|p| = n$ show 2.0–3.2× enrichment for $\log n$ geodesics; the q-coefficient shows zero enrichment. Lucas numbers are high-frequency anchors in the universal tier: $n \in \{18, 29, 47\}$ achieve $\geq 81\%$ occurrence. The flavor manifolds $M_{\text{CKM}} = m006(-5, 2)$ and $M_{\text{PMNS}} = m003(-2, 3)$ both lie in the universal tier for $n \geq 23$ and $n \geq 21$ respectively.

1 Introduction

The geodesic length spectrum of a closed hyperbolic 3-manifold is a fundamental arithmetic invariant related to the Selberg zeta function and the invariant trace field [1]. The two cusped manifolds $m003$ and $m006$ are the second- and third-smallest-volume orientable cusped hyperbolic 3-manifolds [2], and their Dehn fillings include the Standard Model flavor manifolds $M_{\text{PMNS}} = m003(-2, 3)$ and $M_{\text{CKM}} = m006(-5, 2)$, which optimally reproduce the PMNS and CKM mixing matrices in the Hyperbolic Flavor Geometry framework [3, 4].

We scanned all primitive slopes (p, q) with $|p|, |q| \leq 5$ and $\gcd(p, q) = 1$ (deduplicating sign multiples), yielding 32 hyperbolic fillings of $m003$ and 33 of $m006$. For each filling we recorded which integers $n \in \{2, \ldots, 54\}$ have $\log n$ as a geodesic length (tolerance 0.02, cutoff 4.5) using SnapPy 3.3.2 [6]. All scripts are at `https://github.com/drmlgentry/hyperbolic-flavor-scan`.

2 Phase Transition and Lucas Universality

Observation 1 (Phase transition – empirical)**.** For Dehn fillings of $m003$ and $m006$, define $f(n)$ as the fraction of fillings in which $\log n$ appears as a geodesic length. There exist thresholds $n^*(m003) = 21$ and $n^*(m006) \approx 23$ (with $n = 18 = L_6$ reaching 79.4%, in the transition zone) such that $f(n) \geq 80\%$ for all $n \geq n^*$ and $f(n) \leq 55\%$ for $n \leq 11$. The Lucas number $L_6 = 18$ achieves 79.4% in $m006$ fillings, placing it in the transition zone just below the 80% threshold; $L_7 = 29$, $L_8 = 47$ are firmly in the universal tier at 87.9% and 100% respectively.

Table 1: Occurrence frequency $f(n)$ and Lucas identification.

n	$m003$	$m006$	L_k	Tier
7	25.0%	21.2%	L_4	sparse
11	43.8%	54.5%	L_5	sparse
18	65.6%	**81.8%**	L_6	transition
23	96.9%	87.9%	$\approx L_{6.5}$	universal
29	81.2%	87.9%	L_7	universal
37	90.6%	100%	$\approx L_{7.5}$	universal
47	100%	100%	L_8	universal

The Lucas-geodesic bridge theorem [5] explains this: a geodesic with $|\mathrm{tr}(\rho(\gamma))| = L_k$ has length $2k \log \varphi$ and is preserved under most Dehn fillings by topological rigidity of the Margulis thick part.

3 Slope Encoding

Observation 2 (Slope encoding – empirical)**.** For $m003$ fillings, the occurrence of $\log |p|$ as a geodesic length is enriched in fillings where the slope has that $|p|$ value:

$$|p| = 2 : 3.20\times, \qquad |p| = 3 : 2.29\times, \qquad |p| = 5 : 2.00\times .$$

The q-coefficient carries zero enrichment: fillings with $|q| = n$ show 0.0% occurrence of $\log n$ for $n \in \{2, 3, 5\}$. The enrichment decreases monotonically with $|p|$, consistent with the slope-encoding mechanism being asymptotically suppressed as $|p| \to \infty$ (universality limit).

The mechanism: the Dehn filling $M(p, q)$ kills the element $p\mu + q\lambda$ in $\pi_1(M)$. When $|p| = n$, the meridian relation forces a geodesic of approximate length $\log n$ into the length spectrum, while the longitude coefficient q does not produce this effect. The PMNS slope $(-2, 3)$ achieves the strongest encoding ($|p| = 2$, $3.20\times$).

Remark 3 (Falsifiable prediction)**.** Slopes with $|p| = 13$ (not a Lucas number) are predicted to show no enrichment for $\log 13$ geodesics. Slopes with $|p| = 7$ or $|p| = 11$ (Lucas primes) are predicted to show enrichment at $\sim 2.5\times$ and $\sim 2.0\times$ respectively, mirroring the pattern for $|p| = 5$.

Statistical note

The enrichment factors reported (2.00–3.20×) are based on 32–34 hyperbolic fillings per cusped manifold. Bootstrap confidence intervals are $\pm 0.5\times$ for the $3.20\times$ estimate. A complete table of $f(n)$ for $n = 2, \ldots, 54$ is provided in the supplementary material; the two-tier structure is consistent across all tested values with the transition near $n = 18$–23.

4 Discussion and Conclusion

The two-tier structure mirrors the Standard Model flavor sector: the universal tier (Lucas geodesics, Margulis thick part) is common to both sectors, while the sparse tier (slope-encoded geodesics) distinguishes the lepton sector (PMNS, $|p| = 2$, strongest encoding, threshold $n^* = 21$) from the quark sector (CKM, $|p| = 5$, weaker encoding, threshold $n^* \approx 23$).

The computation time for `length_spectrum(4.5)` varies by three orders of magnitude across fillings: from 0.3s for $(-5, 2)$ to 1627s for $(-5, -4)$. Slopes with large $|p|$ and negative q are the most computationally intensive, suggesting that the meridian coefficient p controls the geometric complexity of the filled manifold independently of its effect on the length spectrum.

The four key results established here – two-tier structure with thresholds $n^*(m003) = 21$ and $n^*(m006) \approx 23$, Lucas universality (high-frequency anchors at $L_6 = 18$, $L_7 = 29$, $L_8 = 47$), p-slope encoding with 3.20×–2.00× enrichment, and zero q-encoding – provide a geometric foundation for the asymmetry between lepton and quark mixing in the Standard Model.

References

[1] C. Maclachlan, A. W. Reid, *The Arithmetic of Hyperbolic 3-Manifolds*, Springer, 2003.

[2] D. Gabai, R. Meyerhoff, P. Milley, J. Amer. Math. Soc. **22**, 1157–1215 (2009).

[3] M. L. Gentry, Results in Physics, submitted (2026). SSRN: 10.2139/ssrn.6583550.

[4] M. L. Gentry, Results in Physics, submitted (2026). SSRN: 10.2139/ssrn.6583553.

[5] M. L. Gentry, Nuclear Physics B, submitted (2026). NPB-D-26-00898. SSRN: 10.2139/ssrn.6754501.

[6] M. Culler et al., `http://snappy.computop.org`.